Isotropy types are not counted by coadjoint orbits

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Scope and result. The final clause of the source identifies the number of isotropy types in nonfree reduction with the number of coadjoint orbits. In the standard Hamiltonian setting this is false: the sign action of the compact Lie group C_2 on the connected symplectic plane has two occurring isotropy types in its zero-level quotient, whereas its coadjoint action has one orbit. We prove this necessary-clause counterexample, without assigning values to the source's other unspecified constants or quantization data. The source imposes neither connectedness of the group nor properness of the moment map. Our group is disconnected, and our moment map is not proper. The claim here is expressly confined to the printed scope.

An actual smooth symplectic action. Let $G = C_2 = \{1, s\}$, with $s^2 = 1$, and give it the discrete topology and the zero-dimensional real smooth manifold structure. Let $V = \mathbb{R}^2$ with its usual smooth structure. Define

$$1 \cdot x = x, \qquad s \cdot x = -x, \qquad \omega_x(u, v) = u_0 v_1 - u_1 v_0.$$

Thus $\omega = dx_0 \wedge dx_1$. This is an alternating continuous bilinear form varying smoothly with x. If $\omega_x(u, v) = 0$ for every v, testing $v = (0, 1)$ and $v = (1, 0)$ gives $u_0 = u_1 = 0$, so it is nondegenerate. In the global vector-space chart the exterior derivative on constant vectors is

$$(d\omega)_x(u, v, w) = D\omega_x[u](v, w) - D\omega_x[v](u, w) + D\omega_x[w](u, v).$$

The form is constant, so its actual Fréchet derivative vanishes and $d\omega = 0$. These are the usual symplectic-form conditions [1, §1.3].

The action is jointly smooth: on each open component $\{g\} \times V$ it is the linear map $x \mapsto \varepsilon(g)x$, where $\varepsilon(1) = 1$ and $\varepsilon(s) = -1$. Its derivative is that same linear map, and

$$\omega_{g\cdot x}(D(g\cdot)_x u, D(g\cdot)_x v) = \varepsilon(g)^2 \omega_x(u, v) = \omega_x(u, v).$$

It therefore acts by symplectomorphisms.

Lie algebra, coadjoint action and moment map. The actual tangent Lie algebra $\mathfrak{g} = T_1 G$ is zero-dimensional. It is a zero vector space, so its dual $\mathfrak{g}^*$ is also a singleton. The adjoint map is defined as the derivative at 1 of $h \mapsto ghg^{-1}$; the coadjoint action is its inverse dual. Consequently the quotient of $\mathfrak{g}^*$ by this actual action has exactly one element.

Define $\mu : V \to \mathfrak{g}^*$ by $\mu(x) = 0$. It is smooth and equivariant. For $\xi \in \mathfrak{g}$, the fundamental vector field is the genuine orbit-map derivative

$$\xi_V(x) = D_1(g \mapsto g \cdot x)[\xi].$$

Since $\xi = 0$, linearity of this derivative gives $\xi_V(x) = 0$. Each component $\mu^\xi(x) = \langle \mu(x), \xi \rangle$ is constant, hence

$$d\mu^\xi_x(v) = 0 = \omega_x(\xi_V(x), v).$$

Together with equivariance, this is precisely the moment-map identity [1, Definition 22.1]. In particular, (V, ω, G, μ) is a Hamiltonian G-space, and its zero level is all of V.

The two occurring types in the quotient. The reduced orbit space is the actual quotient

$$M_0 = \mu^{-1}(0)/G = V/G.$$

Its isotropy type at the orbit of x is the conjugacy class of the stabilizer $G_x = \{g : g \cdot x = x\}$. These are the standard orbit-type indices [2, Example 1.2 and Theorem 2.1]. Direct calculation gives

$$G_0 = G, \qquad G_x = \{1\} \quad (x \neq 0),$$

because $-x = x$ over $\mathbb{R}$ implies $x = 0$. Both cases occur, for instance at 0 and $(1, 0)$. The action is therefore nonfree. The two subgroups are distinct and cannot be conjugate; indeed G is abelian, so conjugation fixes every subgroup. Isotropy type is invariant along orbits and therefore defines a function on M_0. Its image is exactly the two-element set $\{[G], [\{1\}]\}$. Thus

$$\#\{\text{occurring isotropy types in } M_0\} = 2 \neq 1 = \#(\mathfrak{g}^*/G).$$

The same mismatch holds if coadjoint orbits are restricted to the image of μ, since that image is already all of the singleton $\mathfrak{g}^*$.

Formal scope. The Lean project uses the native `GroupLieAlgebra`, actual `mfderiv` for the adjoint and fundamental fields, and the continuous dual with its explicit linear equivalence to the algebraic dual. The plane's canonical tangent spaces carry the displayed bilinear form. Smoothness, nondegeneracy, the coordinate exterior derivative, and the moment identity are proved from these definitions. The zero level is a subtype, its reduction is the native `MulAction.orbitRel` quotient, and its isotropy types come from native `MulAction.stabilizer` subgroups modulo genuine conjugacy. Finiteness and nonemptiness are proved before the two cardinalities are compared. No cardinality is stipulated as a definition.

The result concerns the source's final count clause. It does not contradict the classical orbit-type decomposition theorem, nor make a claim about versions with extra connected-group or proper-moment-map hypotheses. The full bilingual source and a pinned Lean 4.33.0 project with Mathlib revision

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are included, together with source-bound verification receipts.

References

[1] A. Cannas da Silva, *Lectures on Symplectic Geometry*, Lecture Notes in Mathematics 1764, Springer, 2001; author version, revised January 2006, pp. 6, 133–135.

[2] R. Sjamaar and E. Lerman, *Stratified symplectic spaces and reduction*, Annals of Mathematics (2) 134 (1991), 375–422; author preprint, Example 1.2 (p. 5), Theorem 2.1 (p. 9).